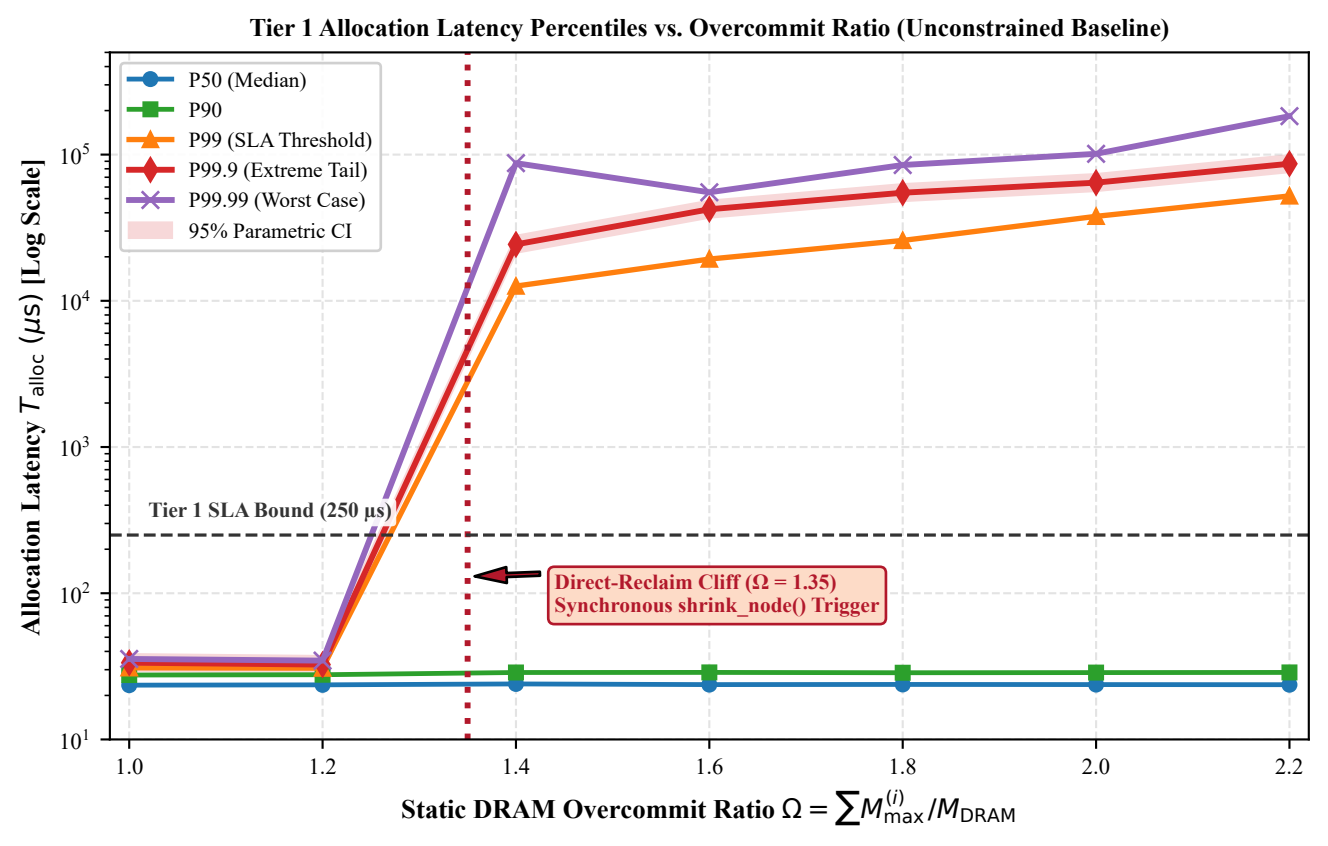

Tier 1 Allocation Latency Percentiles vs. Overcommit Ratio (Unconstrained Baseline)
P50 (Median)
P90
P99 (SLA Threshold)
P99.9 (Extreme Tail)
P99.99 (Worst Case)
95% Parametric CI
Tier 1 SLA Bound (250 µs)
Direct-Reclaim Cliff (Ω = 1.35)
Synchronous shrink_node() Trigger
Allocation Latency $T_{\text{alloc}}$ ($\mu$S) [Log Scale]
$10^1$
$10^2$
$10^3$
$10^4$
$10^5$
1.0
1.2
1.4
1.6
1.8
2.0
2.2
Static DRAM Overcommit Ratio $\Omega = \sum M_{\text{max}}^{(i)} / M_{\text{DRAM}}$